

Section II: Case III

**Defined Contribution Plan and MPF Retirement
Planning**

AMA1D03 Introduction to Pension Mathematics

(a) Descriptions of the Defined Contribution Plan and Its Application in the Pension System

1. Defined Contribution (DC) versus Defined Benefit (DB)

In a defined contribution plan, contributions are set in advance but the retirement benefit is not. The employer and employee pay into a personal account, and the final balance depends on how much was contributed, how investments performed, and what fees were charged. The member carries most of the investment and longevity risk.

In a defined benefit plan, the situation is reversed. The benefit formula is fixed, but the contributions needed to fund it are not. The sponsor uses actuarial assumptions to set contribution levels and bears most of the risk if the plan falls short. Hong Kong also has Occupational Retirement Schemes that can be run on either a DC or DB basis. These sit alongside mandatory MPF as part of the second pillar.

Hong Kong's Mandatory Provident Fund System began on **1 December 2000**. It is a compulsory DC scheme and forms the second pillar of the city's retirement framework:

- **First pillar:** Flat-rate cash benefits subject to means tests. CSSA is the last-resort safety net. OALA targets lower-income seniors. OAA is more widely available but pays less.
- **Second pillar:** Mandatory and voluntary MPF contributions, together with ORSO where it applies.
- **Third pillar:** Personal savings, insurance, and family support.

Hong Kong has no public pension linked to past earnings. First-pillar payments are fixed amounts, so MPF is the main retirement base for the middle class. This is a major difference from systems such as Canada's CPP or US Social Security.

2. Types of MPF Schemes

The MPFA classifies schemes into three types:

Table 1: Types of MPF schemes

Scheme Type		Description
Master Trust Schemes		Open to employees of participating employers and self-employed persons
Employer-sponsored Schemes		Only for employees of a single employer
Industry Schemes		For workers in sectors such as catering and construction

Each employee holds a personal account within a scheme. When changing jobs, accrued benefits can be transferred to a new employer's scheme or kept in a personal account.

3. Mandatory and Voluntary Contributions

Mandatory Contributions:

- Employee and employer each contribute **5%** of relevant income.
- Relevant income has a monthly floor of HK\$7,100 and a ceiling of HK\$30,000.
- If monthly income falls below HK\$7,100, the employee pays nothing, but the employer still contributes 5% of HK\$7,100.
- The maximum mandatory contribution from either party is HK\$1,500 per month.

Voluntary Contributions:

- **In-scheme VC** is paid into the existing MPF plan. It does not qualify for salaries-tax relief and follows the same withdrawal rules as mandatory contributions.
- **TVC** is held in a separate personal account. Up to **HK\$60,000 per year** can be deducted for salaries tax, and the withdrawal rules are more flexible.

In Case III, Peter and Mary each contribute HK\$1,000, which is 5% of a HK\$20,000 salary. Their employers match this amount, so total monthly contributions are **HK\$2,000**.

4. Benefits, Withdrawal, and Retirement Age

Members can normally withdraw accrued benefits at age **65**. Benefits may also be taken early on permanent departure from Hong Kong, total incapacity, terminal illness, or retirement at age 60 after leaving employment. Members with total accrued benefits of HK\$5,000 or less may apply for small-balance withdrawal. On death, benefits are paid to nominated beneficiaries.

At retirement, members may take a lump sum or convert part of the balance into an annuity through the MPF platform. Annuities help address longevity risk, which is one of the main weaknesses of DC plans when members live longer than their savings last.

5. Investment Options, Returns, and Fees

MPF schemes offer Hong Kong, China, Asia, and global equity funds, mixed-asset funds, bond and guaranteed funds, and index funds such as the Tracker Fund of Hong Kong.

Members bear investment risk. Fund managers charge annual fees, usually between 0.5% and 2%, which reduce net returns over long periods. In Case III, Peter invests in a global fund and Mary in a Hong Kong fund. This shows how fund choice affects outcomes even when the contribution pattern is the same.

6. Tax Treatment and System Role

Mandatory MPF contributions are tax-exempt up to statutory limits. TVC gives members an extra tax incentive if they want to close a retirement income gap. Research and industry practice both suggest that mandatory contributions alone are **not enough** to maintain pre-retirement living standards. Members are expected to add voluntary savings and, where they qualify, government cash benefits.

(b) Compare and Analyze the Defined Contribution Plan: Peter and Mary Examples

1. Modelling Setup

We simulate monthly MPF accumulation from December 2000 until each member reaches age 65. The model is written in Python and saved as `case3/main.py`. Table 2 lists the main assumptions.

Notes on the assumptions:

- **Salary growth linked to CPI.** This treats real wages as flat over a member's career. In practice, wages often rise faster than inflation early in a career, which would increase final salary and push the replacement ratio down further.
- **Investment returns.** We use actual monthly returns up to the latest available data, then assume **4% per year** on a nominal basis thereafter. This matches common long-run planning figures used by the MPFA and industry. The model is not a pure historical back-test. Global funds outperformed Hong Kong equities over 2000–2025, but that reflects one historical window and should not be read as a forecast.
- **4% withdrawal rule.** Retirement income in the first year equals 4% of the account balance, with later withdrawals assumed to rise with inflation. The rule was developed for 30-year retirements in the United States. Hong Kong residents often live beyond 85, so a fixed withdrawal rate can leave members exposed to running out of money.
- **Fees.** We assume 1% per year, which sits in the middle of the MPF range. Index funds often charge less than 0.5%, while active equity funds commonly charge between 1.2% and 1.8%. Over 25 to 45 years, even small fee differences matter.

Table 2: Key modelling assumptions

Parameter	Value	Note
Start date	1 Dec 2000	MPF launch
Retirement age	65	Normal retirement
Initial monthly salary	HK\$20,000	Case III
Monthly MC, employee plus employer	HK\$2,000	5% + 5%
Salary growth	HK Composite CPI	Real wages held constant
Hong Kong Fund proxy	Hang Seng Index, $\hat{HSI}$	Tracker Fund tracks HSI
Global Fund proxy	Spliced $\hat{GSPC}$, EFA, ACWI	History from Dec 2000
Forward return	4% p.a., nominal	After latest market data
Fund expense ratio	1% p.a.	Mid-range MPF fee
Retirement income	4% withdrawal rule	First-year income equals 4% of balance

Table 3: Case III participant settings

	Peter	Mary
Start age, Dec 2000	20	40
Retirement age	65	65
Contribution period	45 years	25 years
Retirement year	2045	2025
Assigned fund	Global, spliced	Hong Kong, ^HSI

2. Baseline Results

Table 4: Baseline simulation results

Indicator	Peter, Global	Mary, Hong Kong
Contribution years	45.1	25.1
Total contributions, HK\$	1,443,931	723,931
Investment gains, HK\$	3,552,431	126,930
Final MPF balance, HK\$	4,996,362	850,862
Final annual salary, HK\$	555,156	374,107
Annual retirement income, HK\$	199,854	34,034
Monthly retirement income, HK\$	16,655	2,836
Replacement ratio	36.0%	9.1%

Figure 1 shows Peter’s balance rising steadily over 45 years. Mary’s balance grows more slowly over 25 years and ends at roughly one-sixth of Peter’s level. Peter’s investment gains of HK\$3.55 million exceed his total contributions of HK\$1.44 million, which illustrates how compounding works in a DC plan over a long horizon.

Hong Kong equities were volatile during Mary’s contribution period from 2000 to 2025, including the post-2000 downturn and the 2008 financial crisis. That helps explain why her final balance is lower than estimates based on incomplete return data.

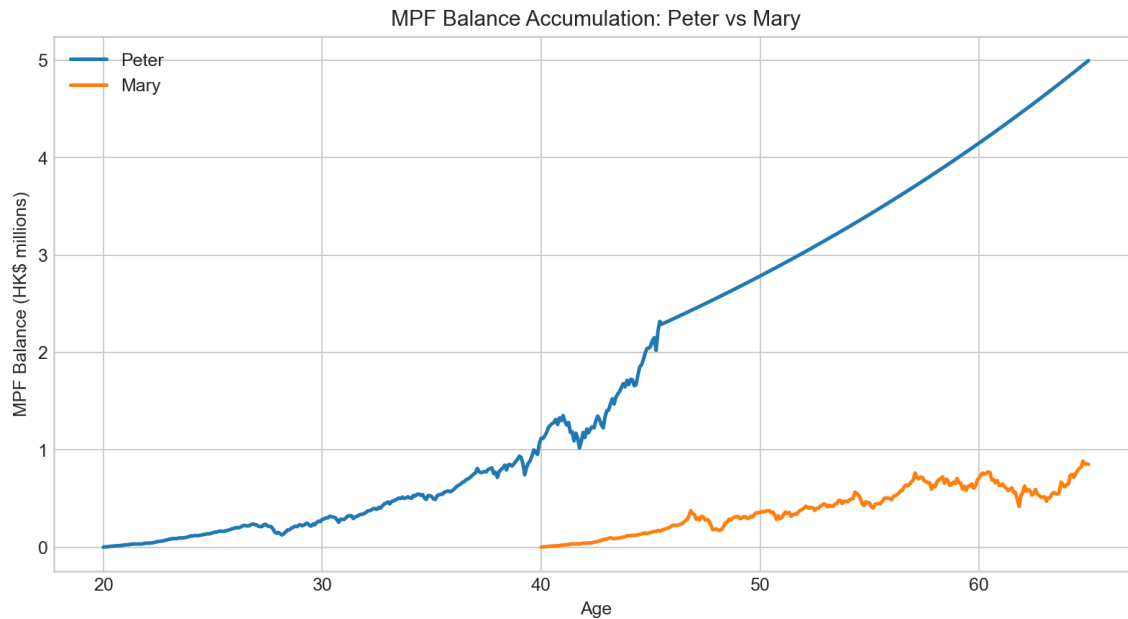


Figure 1: MPF balance accumulation: Peter versus Mary

3. Advantages and Disadvantages

Advantages:

1. **Individual ownership and portability.** Accounts belong to the member and move with them, which suits Hong Kong's mobile labour market. There is no employer credit risk of the kind seen in DB plans.
2. **Employer matching.** The employer's 5% contribution doubles savings without extra cost to the employee.
3. **Investment choice.** Members can choose funds that match their risk tolerance.
4. **Tax relief on TVC.** Up to HK\$60,000 of TVC per year is tax-deductible.
5. **Long compounding horizons.** Members who start early benefit most. At 4% per year, the Rule of 72 suggests wealth doubles about every 18 years. Over 45 years that compounds to roughly six times the original sum, compared with about 2.7 times over 25 years.

Disadvantages:

1. **Market risk and sequence risk.** Members bear the full downside. A sharp fall in markets just before retirement can cut the final balance sharply.
2. **No pooling of risk.** Unlike DB plans, DC plans do not share longevity or investment short-falls across members.
3. **Low replacement ratios.** Even Peter reaches only 36% on mandatory contributions, below the 60–70% range often used in financial planning.

4. **Fees and poor decisions.** Management fees compound against the member over time, and frequent fund switching often reduces returns further.
5. **Penalty for late starters.** Mary's 9.1% replacement ratio shows that beginning at age 40 cannot be fixed simply by picking a better fund.

4. Start Age versus Fund Choice

Table 5 reports replacement ratios by start age. Table 6 shows what happens when Peter and Mary swap funds.

Table 5: Replacement ratio by start age and fund

Start Age	Fund	Final Balance, HK\$	Replacement Ratio
20	Global	4,996,362	36.0%
20	Hong Kong	2,427,886	17.5%
30	Global	3,417,078	30.0%
30	Hong Kong	1,500,213	13.2%
40	Global	2,063,587	22.1%
40	Hong Kong	850,862	9.1%
50	Global	520,922	6.9%
50	Hong Kong	457,617	6.0%

Table 6: Fund-swap sensitivity

Name	Assigned RR	Swapped RR	Change, pp
Peter, age 20	36.0%	17.5%	-18.5
Mary, age 40	9.1%	22.1%	+13.0

Key finding: For the same fund, starting at age 20 rather than age 40 widens the replacement ratio by 27 percentage points, from 36.0% to 9.1% for the global and Hong Kong cases shown above. Peter at age 20 with the weaker Hong Kong fund still ends with a much higher balance than Mary. **How long you contribute matters more than which fund you pick**, because terminal value responds more strongly to time than to return.

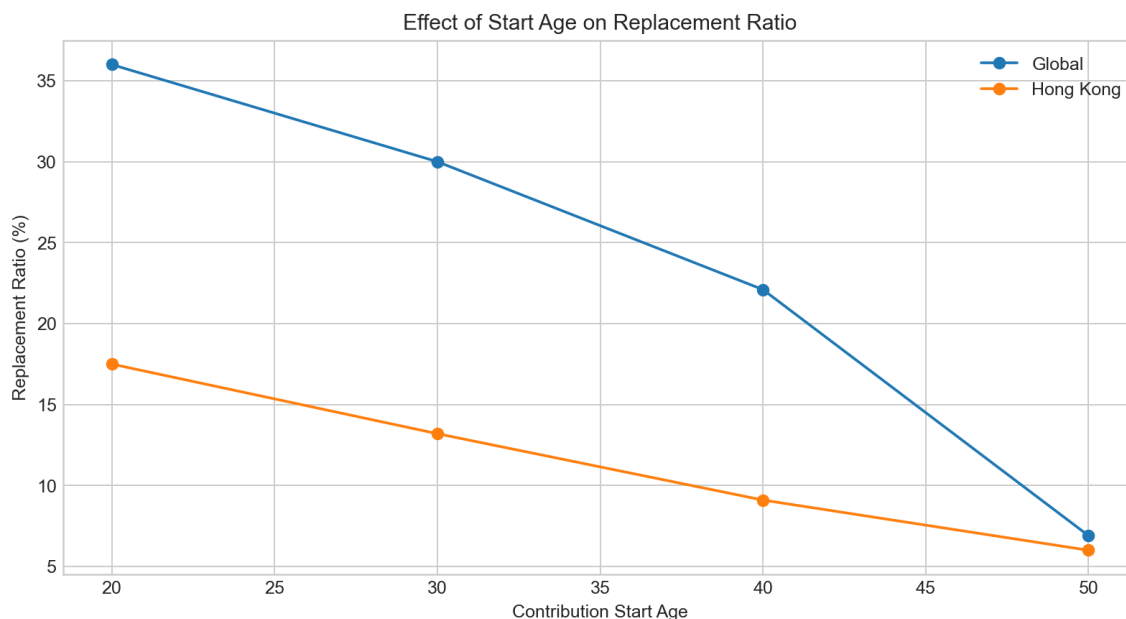


Figure 2: Effect of start age on replacement ratio

5. Living Expenses, Inflation, and Extra Contributions

Many planners aim for a **60–70% replacement ratio**. Using final monthly salary from Table 4, the monthly gap to 60% is about HK\$11,103 for Peter and HK\$15,870 for Mary.

Salaries and mandatory contributions move with CPI, but the real value of retirement income still depends on whether investment returns beat inflation. In a fee sensitivity check, Peter's balance falls from HK\$4.99 million at a 1% fee to about HK\$4.32 million at 1.5% over 45 years.

Table 7: TVC required to reach target replacement ratio

Member	Current RR	Target RR	Extra Monthly TVC, HK\$
Peter	36.0%	60%	1,582
Peter	36.0%	70%	2,242
Mary	9.1%	60%	13,205
Mary	9.1%	70%	15,799

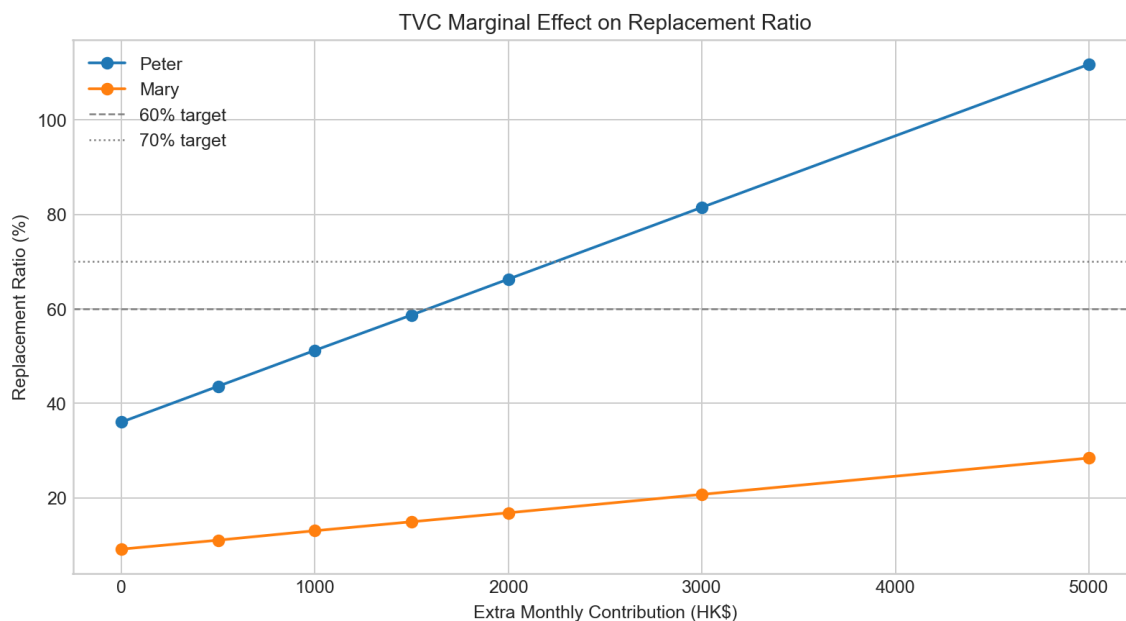


Figure 3: Marginal effect of TVC on replacement ratio

Mary needs large extra contributions or other income sources. Peter needs about HK\$1,600 more per month to reach 60%, which fits within the TVC tax limit of HK\$60,000 per year, or about HK\$5,000 per month. Mary would need HK\$13,205 per month, far above that cap. Only the first HK\$5,000 per month would receive tax relief. The rest would have to come from non-deductible VC or third-pillar savings.

(c) Supporting Retirement Under DC with the Social Security System

1. Hong Kong Three-Pillar Framework

Unlike Canada's CPP or US Social Security, Hong Kong has no universal pension tied to earnings. Government support is means-tested and works in layers:

- The **first pillar** gives a basic floor to seniors with low assets and low income. CSSA has the strictest tests, OALA is subject to asset and income limits, and OAA is easier to qualify for but pays less.
- The **second pillar** is mandatory MPF for employed workers.
- The **third pillar** is where most people must look if they want to keep their standard of living in retirement.

For middle-income workers like Peter and Mary, first-pillar benefits are largely ruled out once MPF balances count toward the asset test. Retirement income has to come from MPF and voluntary savings, not from stacking pillars on top of one another.

2. Old Age Living Allowance

Hong Kong residents aged 65 or above may receive Normal OALA of HK\$2,065 per month or Higher OALA of HK\$4,185 per month in 2025/26, provided they pass income and asset tests. For a single person, the asset limit for both tiers is about HK\$401,000. **MPF benefits that have not yet been withdrawn count as assets.** The more a member saves in a DC plan, the less likely they are to receive OALA. The two sources of support do not simply add together.

3. Combined Retirement Income

Table 8 shows the result after the asset test. Peter's MPF balance is HK\$4.99 million and Mary's is HK\$851,000. Both are above the limit, so neither receives OALA when MPF is counted. Table 9 keeps the old OALA figures only as a hypothetical comparison, assuming assets were below the threshold. That assumption does not apply to Case III.

Table 8: Retirement income after OALA asset test, MPF counted

Member	MPF Balance	Asset Limit	MPF Income	OALA	Combined RR
Peter	4,996,362	401,000	16,655	0	36.0%
Mary	850,862	401,000	2,836	0	9.1%

Table 9: Hypothetical OALA if MPF were excluded from assets, not applicable to Case III

Member	Scenario	MPF	OALA	Total	Combined RR
Peter	Normal OALA	16,655	2,065	18,720	40.5%
Peter	Higher OALA	16,655	4,185	20,840	45.0%
Mary	Normal OALA	2,836	2,065	4,901	15.7%
Mary	Higher OALA	2,836	4,185	7,021	22.5%

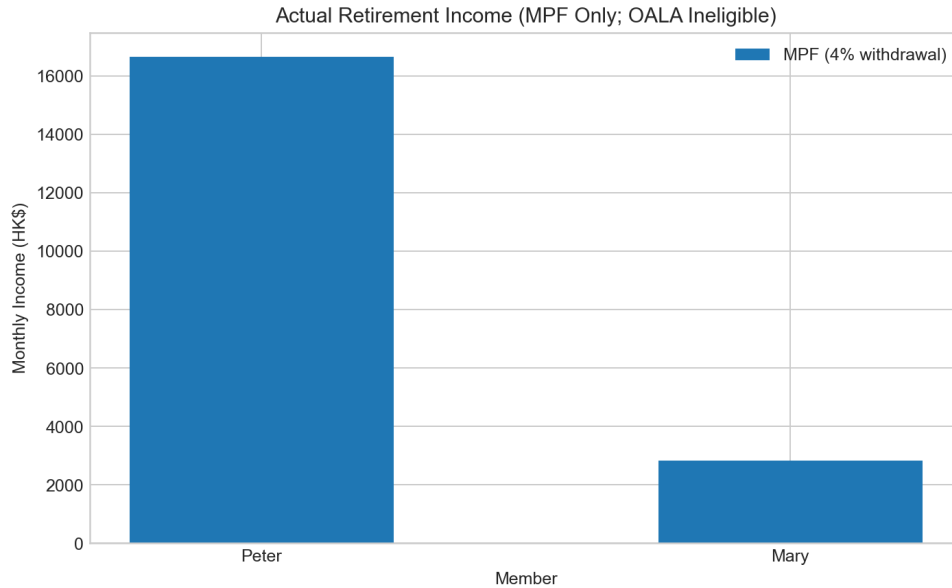


Figure 4: Actual retirement income from MPF; OALA not payable under the asset test

4. Illustrative Strategies

Peter, retiring in 2045:

- MPF already gives a reasonable base at 36%, but he will not qualify for OALA given his expected MPF assets.
- Adding about HK\$1,600 of TVC per month would bring MPF income up to a 60% replacement ratio within the tax-deductible TVC limit.
- A lifecycle approach makes more sense than holding global equity for all 45 years. Higher equity exposure early on, with a gradual shift toward bonds and capital-preservation funds after age 55, would reduce sequence risk.
- Converting part of the MPF balance into an annuity would provide a guaranteed income floor and help manage longevity risk.

Mary, retiring in 2025:

- Mandatory MPF alone leaves her at 9.1%. Her balance of HK\$851,000 is above the OALA asset limit of HK\$401,000, so she will not receive OALA either.
- Working until age 70 raises her replacement ratio only to 10.9% in our simulation. Extra years of contributions help, but they cannot make up for starting 20 years later than Peter.
- The savings she needs are far above the TVC tax cap. She would need non-deductible VC, third-pillar savings, and possibly part-time work.
- With a smaller account, annuitising part of the balance or delaying withdrawals could still improve income security.

A workable plan for Mary might combine MPF income of HK\$2,836 with third-pillar savings or part-time work of about HK\$5,000 per month, giving roughly HK\$7,836 per month. That remains below the 60% target of HK\$18,706. Even in the hypothetical Higher OALA case in Table 9, she would reach only 22.5%.

Table 10: Mary: delayed retirement sensitivity

Retirement Age	Contribution Years	MPF Balance, HK\$	Replacement Ratio
65	25.1	850,862	9.1%
67	27.1	930,635	9.6%
70	30.1	1,128,770	10.9%

5. Conclusions on Extra Contributions

Table 11: Summary recommendations

Question	Peter	Mary
Is mandatory MPF sufficient?	No, 36%	No, 9.1%
Should they add TVC?	Yes, moderate amount	Yes, urgently
Can social security fill the gap?	No, asset test	No, asset test

Hong Kong's DC system places most retirement responsibility on the individual. Peter's case shows that starting early and contributing steadily is the strongest lever available. OALA helps low-asset retirees, but meaningful MPF balances crowd it out for the middle class. **Both Peter and Mary need extra contributions. Mary faces a much larger gap and far less time to close it.**

6. Core Conclusions

1. Mandatory MPF alone is not enough to protect pre-retirement living standards. Voluntary savings are necessary.
2. **When contributions begin** matters more than fund choice in a DC plan, because time drives compounding more than return differences do.
3. Hong Kong's first pillar is a safety net for low-asset retirees, not a top-up for middle-income members with substantial MPF balances.
4. A sound retirement plan combines mandatory MPF, TVC within tax limits, third-pillar savings, gradual de-risking near retirement, and partial annuitisation.

Key Formulae

Future value with CPI-linked contributions, monthly simulation:

$$FV = \sum_{t=1}^n PMT_0(1+g)^{t-1} \prod_{k=t+1}^n (1+r_k) \quad (1)$$

where PMT_0 is the initial monthly contribution, g is the annual growth rate of contributions, r_k is the net return in period k , and n is the number of contribution periods.

When r and g are constant and $r \neq g$, the closed form for a growing annuity is:

$$FV = PMT_0(1+r) \cdot \frac{(1+r)^n - (1+g)^n}{r-g} \quad (2)$$

Replacement ratio under a 4% withdrawal rate:

$$RR = \frac{FV \times 0.04}{Salary_{final}} \quad (3)$$

To find the monthly TVC needed for a target replacement ratio ρ^* , solve for PMT_{TVC} in

$$\rho^* = \frac{0.04 \times FV(PMT_0, g, r, n) + 0.04 \times FV(PMT_{TVC}, g, r, n')}{Salary_{final}} \quad (4)$$

where n' is the number of periods over which extra contributions are paid. In this case, n' equals the full contribution horizon. Table 7 is produced by binary search in the simulation model.

Monthly income gap to a 60% target:

$$Gap_{monthly} = 0.60 \times \frac{Salary_{final}}{12} - \frac{FV_{MPF} \times 0.04}{12} \quad (5)$$

References

- [1] Mandatory Provident Fund Schemes Authority, *MPF Fund Performance Platform*, <https://mfp.mpfa.org.hk/>.
- [2] Social Welfare Department, *Old Age Living Allowance*, <https://www.swd.gov.hk/oala/>.
- [3] Census and Statistics Department, *Composite Consumer Price Index*.
- [4] Investopedia, *Individual Retirement Account*, <https://www.investopedia.com/terms/i/ira.asp>.